

1. Administrative & Study Identification

Full Study Title: A Phase 2, Open-label, Randomized Trial to Evaluate Two Dosing Regimens of Subcutaneous Formulation of Nivolumab in Combination With Intravenous Ipilimumab and Chemotherapy in Participants With Previously Untreated Metastatic or Recurrent NSCLC **Short Title/Acronym:** CheckMate-1533 **Protocol Number:** CA209-1533 **NCT Number:** NCT06946797 **Posting ID:** 380139453263520563 **Sponsor/Company Name:** Bristol-Myers Squibb **Trial Status:** RECRUITING (Currently recruiting participants) **Investigational Product:** Subcutaneous Nivolumab, Intravenous Ipilimumab, and Chemotherapy (Carboplatin, Pemetrexed, Cisplatin [Trade Name: Platinol, ATC Code: L01XA01], Paclitaxel) **Phase of Development:** Phase 2 **Medical Monitor:** Bristol-Myers Squibb Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the pharmacokinetics (including maximum observed serum concentration (Cmax), time to peak concentration (Tmax), and area under the concentration-time curve within a dosing interval (AUC(TAU))) of subcutaneous Nivolumab when administered in two different dosing regimens in combination with intravenous Ipilimumab and chemotherapy. * **Secondary Objectives:** To evaluate the safety, tolerability, and immunogenicity (development of anti-nivolumab and anti-ipilimumab antibodies) of the treatment regimens over a period of up to 2.5 years, including number of participants with adverse events (AEs), serious adverse events (SAEs), and drug-related AEs.

2.2 Background & Rationale To address the clinical need for more accessible and convenient dosing options by evaluating the pharmacokinetics and safety of a subcutaneous formulation of Nivolumab. Providing a subcutaneous option could reduce administration time and healthcare resource utilization compared to traditional intravenous administration for participants with previously untreated metastatic or recurrent NSCLC.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (comparing two investigational regimens) * **Blinding:** Open-label * **Randomization:** Randomized * **Trial Status:** RECRUITING

3.2 Planned Enrollment & Duration * Target \$\$\$: 76 participants *
Number of Sites: Global multicenter trial (includes sites in the US, Brazil, Chile, Mexico, Turkey, France, Italy, Romania, Poland, and Greece) * **Estimated Study Start:** 19-Sep-2025 *
Estimated Primary Completion: 05-Feb-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed stage IV or recurrent non-small cell lung cancer (NSCLC) (as defined by the 9th edition of the IASLC Lung Cancer Staging Guidelines) of squamous or non-squamous histology. 2. No prior systemic anti-cancer treatment (including EGFR, ALK, ROS-1, BRAF, RET, and NTRK inhibitors) given as primary therapy for advanced or metastatic disease. 3. Eastern Cooperative Oncology Group (ECOG) Performance Status ≤ 1 at screening and confirmed prior to randomization. 4. Measurable disease by CT or MRI per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 criteria with radiographic tumor assessment performed within 28 days of randomization. 5. Prior definitive chemoradiation for locally advanced disease is permitted as long as the last administration of chemotherapy or radiotherapy occurred at least 6 months prior to randomization. Participants with locally advanced disease with recurrence after chemoradiation therapy (stage III disease with no curative options) are eligible. 6. Prior adjuvant or neoadjuvant chemotherapy for early-stage lung cancer is permitted if completed at least 6 months prior to randomization.

4.2 Exclusion Criteria 1. Any active, known, or suspected autoimmune disease (exceptions apply for conditions like type I diabetes, hypothyroidism on hormone replacement, skin disorders not requiring systemic treatment, or conditions not expected to recur without an external trigger). 2. Any prior treatment with an anti-PD-1, anti-PD-L1, anti-PD-L2, anti-CTLA-4 antibody, or any other drug specifically targeting T-cell co-stimulation or checkpoint pathways. 3. Known driver mutations with available targeted therapy (including but not limited to EGFR mutations, ALK translocations, ROS-1 translocations, known BRAFV600E, and activating RET mutations and NTRK fusion gene alterations). 4. Presence of leptomeningeal metastases (carcinomatous meningitis). 5. Untreated central nervous system (CNS) metastases. 6. Previous malignancies (except non-melanoma skin cancers, and in situ cancers such as bladder, gastric, colon, cervical/dysplasia, melanoma, or breast) unless a complete remission was achieved at least 2 years prior to randomization and no additional therapy is required or anticipated. 7. History of interstitial lung disease or pneumonitis that required oral or IV glucocorticoids to assist with management.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * Dosage/Frequency: Participants are randomized to receive one of two distinct dosing regimens of subcutaneous Nivolumab alongside intravenous Ipilimumab and platinum-based chemotherapy. *** Route of Administration:** Subcutaneous (Nivolumab) and Intravenous (Ipilimumab, Carboplatin, Pemetrexed, Cisplatin [Platinol], Paclitaxel). *** Duration of Treatment:** Maximum treatment/follow-up duration up to approximately 2.5 years for safety evaluations.

5.2 Concomitant & Prohibited Therapy * Permitted: Inhaled or topical steroids, and adrenal replacement steroid ≤ 10 mg daily prednisone equivalent in the absence of active autoimmune disease. *** Prohibited:** Systemic treatment with corticosteroids (>10 mg daily prednisone equivalent) within 14 days or other immunosuppressive medications within 30 days of randomization.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, ECOG assessment, CT/MRI RECIST 1.1 tumor assessment
Baseline	Day 0	Randomization, First Dose
Interim	Weeks 1 - 18	PK sampling up to 3 weeks (C _{max} , T _{max} , AUC), safety labs, C _{trough} at Cycle 7 Day 1 (Week 18)
End of Study	Up to 2.5 Years	AE/SAE monitoring, final survival follow-up, and IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * Definition: Pharmacokinetic (PK) parameters of subcutaneous Nivolumab in serum up to 3 weeks, primarily Maximum observed serum concentration (C_{max}), Time to peak concentration (T_{max}), and Area under the concentration-time curve within a dosing interval (AUC(TAU)). *** Measurement Method:** Serum PK concentration blood sampling analysis.

7.2 Secondary & Exploratory Endpoints * Incidence of Adverse Events (AEs), Serious Adverse Events (SAEs), and drug-related AEs over approximately 2.5 years. * Incidence of Immune-Mediated Adverse Events (IMAEs) and number of deaths. * Immunogenicity measured by the number of participants developing anti-nivolumab and anti-ipilimumab antibodies.

8. Safety & Adverse Event Reporting

- Adverse Event (AE) Monitoring:** Continuous collection of overall AEs, drug-related AEs, and specific IMAEs via

clinical visits and laboratory findings over a period of up to 2.5 years.

- **Serious Adverse Events (SAEs):** Tracked through routine pharmacovigilance and reporting standards up to approximately 2.5 years.
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee appointed to oversee safety thresholds and protocol integrity.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Pharmacokinetic (PK) Set for primary endpoints; Safety Set (all randomized patients who received ≥ 1 dose) for safety events.
 - **Sample Size Justification:** $N=76$, estimated to provide adequate statistical power to evaluate the PK exposures of the two subcutaneous dosing regimens.
 - **Handling of Missing Data:** Executed according to standard operating procedures (e.g., LOCF or mixed-model repeated measures) for oncology PK studies.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Ongoing onsite and remote monitoring to ensure RECIST 1.1 evaluation compliance, PK sampling timeline adherence, and reporting of IMAEs.
 - **Data Source Level:** 5
 - **Audit Trail:** Standardized secure eCRF entry, locking, and data clarification mechanisms per Bristol-Myers Squibb guidelines.
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11. Study Identifiers & Governance

- **Primary ID:** CA209-1533
- **Registry Number:** NCT06946797
- **Posting ID:** 380139453263520563
- **Sponsor/Lead Organization:** Bristol-Myers Squibb
- **Study Title:** A Study to Evaluate Two Dosing Regimens of Subcutaneous Nivolumab in Combination With Intravenous Ipilimumab and Chemotherapy in Participants With Previously Untreated Metastatic or Recurrent Non-Small Cell Lung Cancer (NSCLC)
- **Official Regulatory Title:** A Phase 2, Open-label, Randomized Trial to Evaluate Two Dosing Regimens of Subcutaneous Formulation of Nivolumab in Combination With Intravenous

12. Clinical Framework (Oncology Specific)

- **Disease Area:** Non-Small Cell Lung Cancer
 - **Primary Condition:** Metastatic or Recurrent Non-Small Cell Lung Cancer (NSCLC)
 - **Preferred UMLS Name:** Non-Small Cell Lung Carcinoma
 - **Diagnosis Threshold:** Stage IV or recurrent NSCLC per the 9th edition of the IASLC Guidelines
 - **Medical Methodology:** First-line systemic anti-cancer therapy combining checkpoint inhibitors and platinum-based chemotherapy
 - **Optimization Target:** Evaluation of Pharmacokinetics and tolerability of subcutaneous immunotherapy administration
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13. Study Procedures & Methodology

- **Management Pathway:** Advanced/metastatic NSCLC first-line treatment protocols
 - **Education Protocol (HCP):** Site initiation and training regarding subcutaneous injection procedures and exact PK draw schedules.
 - **Education Protocol (Patient):** Education on immune-mediated toxicities reporting and treatment scheduling.
 - **Quality Audit Mechanism:** Real-time RECIST 1.1 tumor assessment auditing and PK assay quality control.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to approximately 2.5 years
 - **Onsite Visit Cadence:** Pre-treatment screening, intensive PK sampling at baseline/interim weeks (up to 3 weeks), and Cycle 7 Day 1 (Week 18)
 - **Remote Monitoring Frequency:** Regular safety tele-checks as dictated by the 2.5-year AE reporting window
 - **Checklist Review Interval:** Disease assessment (CT/MRI) starting at 28 days post-randomization and at standard protocol-defined intervals thereafter.
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15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator			[Insert Name]		_____				[DD-MMM-YYYY]							
	Clinical Operations Lead			[Insert Name]		_____				[DD-MMM-YYYY]							
	Lead Statistician			[Insert Name]		_____				[DD-MMM-YYYY]							